

Multi-centric Geometry and Discrete Harmonic Structures: Analytical and Differential Foundations of the Thutmos Transformation (TTH)

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Abstract

This article introduces and formalizes a new geometric paradigm known as the Thutmos Transformation (TTH). In contrast to the global rigid transformations of classical Euclidean geometry, the TTH models space through a multicentric deformation process. In its discrete and affine forms, we demonstrate that the TTH generates a dynamical system that obeys the Banach fixed-point theorem, and that its extension to regular polygons reveals a strict isomorphism with the Discrete Fourier Transform (DFT), thereby establishing a discrete harmonic geometry. More fundamentally, the generalization of TTH through the introduction of spatial weighting functions (partition of unity) leads to a loss of linearity and commutativity. The differential analysis of this operator highlights the endogenous emergence of a non-trivial Riemannian metric (g_{ij}) , generating a spatial curvature linked to the deformation gradients. These results bridge the gap between affine algebra, spectral analysis, and differential geometry.

Keywords: Multi-centric transformation, Riemannian Geometry, Discrete Harmonic Analysis, Affine dynamical systems, Induced metric, Lie Algebra, Geometric Fourier Transform.

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Contents

1	Introduction	4
2	Geometric Genesis: The Discrete Euclidean Case	4
2.1	Initial construction and determination of foci	4
2.2	Formal definition of the elementary constant-weight TTH . . .	5
2.3	Immediate affine properties	5
3	Spectral and Dynamic Analysis	6
3.1	Study of eigenvalues	7
3.2	Study of asymptotic stability	7
3.3	Topological classification of the dynamical system	8
4	Discrete Harmonic Structure: Extension to Polygons	8
4.1	Definition via trigonometric projections	8
4.2	The case of the square ($n = 4$): quadratic invariance	9
4.3	Fundamental theorem and Fourier isomorphism	9
5	Algebra of Multi-centric Transformations	10
5.1	Definition of the transformation space $\mathcal{T}$	10
5.2	Composition law and formal non-commutativity	11
5.3	Lie algebra type structure	11
6	Differential Geometry and Induced Metric	11
6.1	Differentiability and Jacobian calculation	11
6.2	Emergence of non-linearity	12
6.3	The induced metric tensor g_{ij}	12
6.4	Calculation of generated curvatures	13
7	Three-dimensional Extensions and Physical Interpretations	13
7.1	Action of the TTH on solids and curvature concentration . . .	13
7.2	Fiber structure of the generated space	14
7.3	Analogy with general relativity	14
8	Conclusion and Perspectives	15
8.1	Synthesis: Discrete harmonic geometry	15
8.2	Open problems and future developments	15

A	Appendix: Graphical Visualizations and Proofs of Concept (PoC)	16
A.0.1	Test PyTTH Affine PoC.py	16
A.0.2	Test PyTTH Diff PoC.py	17
A.0.3	PyTTH ML Classifier.py	18
A.0.4	PyTTH Neural.py	19
A.0.5	PyTTH Harmonic Fourier.py	20
A.0.6	PyTTH Fourier Filter.py	21
A.0.7	PyTTH Spectra.py	22
A.0.8	PyTTH Riemann.py	23
A.0.9	PyTTH Lie Algebra.py	24
A.0.10	PyTTH Topology 3D.py	25

1 Introduction

Classical geometry traditionally revolves around the study of invariances under the action of rigid, affine, projective, or conformal transformation groups. In these standard frameworks, space is often perceived as a passive receptacle whose metric and topology are fixed *a priori*. In this article, we introduce a new transformational paradigm, called the **Thutmos Transformation (TTH)**, which substitutes classical global maps with a multi-centric deformation process. The fundamental intuition behind this theory rests on a seemingly simple geometric problem, initially formulated in a purely visual and kinematic manner: how to "walk" a Euclidean figure (such as a right-angled triangle) on multiple circles simultaneously, so that its trajectory generates a multiplicity of images parameterized by a combined "circular motion"? The synthetic resolution of this problem highlights the necessity of coupling multiple centers of rotation. By axiomatizing this construction, we demonstrate that the Thutmos Transformation is not merely a kinematic superposition, but a true mathematical operator. The main results of this article show that the TTH, in its discrete formulation, generates a continuous affine dynamical system exhibiting remarkable properties such as barycentric invariance. More fundamentally, we prove that when the TTH is generalized through spatially dependent weighting functions, it loses its linearity to locally induce a non-trivial Riemannian metric, thus bridging an unexpected gap between elementary Euclidean geometry, discrete harmonic analysis, and differential geometry (with strong analogies to general relativity).

2 Geometric Genesis: The Discrete Euclidean Case

The emergence of the Thutmos Transformation originates in a straightedge and compass construction, before being abstracted into the formalism of Euclidean affine spaces.

2.1 Initial construction and determination of foci

Let us consider a right-angled triangle $\Delta = ABC$ in the Euclidean plane. The starting postulate of the TTH consists of determining "action centers"

exterior to the figure by completing parallelograms. Algorithmically, we determine the fourth points forming parallelograms with the vertex triplets (A, B, C) , (C, A, B) , and (A, C, B) . Let O_2 , O_3 , and O_1 be these respective points.

From this configuration, we draw rays originating from the original vertices passing through these new centers, and we define the distances $R_2 = \|BO_2\|$, $R_3 = \|AO_3\|$, and $R_1 = \|CO_1\|$. These distances constitute the respective radii of a family of circles $\mathcal{C}_1(O_1, R_1)$, $\mathcal{C}_2(O_2, R_2)$, and $\mathcal{C}_3(O_3, R_3)$. The propagation of the triangle ABC on these three circular orbits generates image triplets (A'_k, B'_k, C'_k) corresponding to discrete evaluations on the trigonometric circle. It is this synchronization of trajectories that calls for a continuous analytical formalization.

2.2 Formal definition of the elementary constant-weight TTH

To systematize this continuous motion, we define the Thutmos Transformation as a convex combination of excentric rotations.

Definition 1 (Elementary TTH). Let n centers $O_1, O_2, \dots, O_n \in \mathbb{R}^2$ (with $n \geq 3$). Let $w_1, w_2, \dots, w_n \in \mathbb{R}_+^*$ be strictly positive weights such that the normalization condition $\sum_{i=1}^n w_i = 1$ is satisfied. Let finally $\phi_1, \dots, \phi_n \in \mathbb{R}$ be initial angular phases. For a continuous sweep angle $\theta \in \mathbb{R}$, the Thutmos Transformation $\mathcal{T}_\theta : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is defined for any point P by:

$$\mathcal{T}_\theta(P) = \sum_{i=1}^n w_i \left(O_i + \mathcal{R}(\theta + \phi_i)(P - O_i) \right) \quad (1)$$

where $\mathcal{R}(\alpha) \in SO(2)$ denotes the rotation matrix by angle α .

In the canonical case derived from the triangular construction, we have $n = 3$ and the transformation superimposes the simultaneous action of three rotation operators anchored at O_1, O_2 , and O_3 .

2.3 Immediate affine properties

Although defined by angular dynamics, the elementary TTH retains an affine nature for any fixed parameter θ .

Proposition 1 (Affine form of the TTH). *For any $\theta \in \mathbb{R}$, the operator $\mathcal{T}_\theta$ is an affine transformation of $\mathbb{R}^2$ admitting the canonical decomposition:*

$$\mathcal{T}_\theta(P) = A(\theta)P + b(\theta) \quad (2)$$

where $A(\theta) \in \mathcal{L}(\mathbb{R}^2)$ is the linear part and $b(\theta) \in \mathbb{R}^2$ is the translation vector.

Proof. Let $P \in \mathbb{R}^2$. By expanding the expression from Definition 1 and exploiting the linearity of the vector space, we isolate the P -dependent terms:

$$\begin{aligned} \mathcal{T}_\theta(P) &= \sum_{i=1}^n w_i O_i + \sum_{i=1}^n w_i \mathcal{R}(\theta + \phi_i) P - \sum_{i=1}^n w_i \mathcal{R}(\theta + \phi_i) O_i \\ &= \left(\sum_{i=1}^n w_i \mathcal{R}(\theta + \phi_i) \right) P + \sum_{i=1}^n w_i \left(I - \mathcal{R}(\theta + \phi_i) \right) O_i \end{aligned}$$

By direct identification, we obtain the matrix linear operator:

$$A(\theta) = \sum_{i=1}^n w_i \mathcal{R}(\theta + \phi_i) \quad (3)$$

and the translation vector, which encodes the perturbed barycentric influence of the foci O_i :

$$b(\theta) = \sum_{i=1}^n w_i \left(I - \mathcal{R}(\theta + \phi_i) \right) O_i \quad (4)$$

□

This algebraic formulation shows that the matrix $A(\theta)$ is generally no longer a simple rotation matrix in $SO(2)$, although it is a convex combination of orthogonal rotation matrices. This loss of strict orthogonality introduces contraction and dilation effects, whose spectral study will define the dynamic behavior of the system.

3 Spectral and Dynamic Analysis

The operator $\mathcal{T}_\theta$ having been identified as an affine transformation with linear part $A(\theta)$, its iterative behavior is entirely dictated by the spectral properties of this matrix. To this end, we exploit the canonical isomorphism between the Euclidean plane $\mathbb{R}^2$ and the complex plane $\mathbb{C}$.

3.1 Study of eigenvalues

Under the isomorphism $(x, y) \mapsto z = x + iy$, the action of an orthogonal rotation matrix $\mathcal{R}(\alpha) \in SO(2)$ on a vector uniquely corresponds to multiplication by the complex phasor $e^{i\alpha}$.

Theorem 1 (Spectrum of the linear operator). *Let $\mathcal{T}_\theta$ be the elementary Thutmos Transformation. Under its complex representation, the associated linear operator $A(\theta)$ admits a single eigenvalue $\lambda(\theta) \in \mathbb{C}$ given by:*

$$\lambda(\theta) = \sum_{j=1}^n w_j e^{i(\theta+\phi_j)} = e^{i\theta} \sum_{j=1}^n w_j e^{i\phi_j} \quad (5)$$

The modulus $|\lambda(\theta)|$ defines the global homothety factor of the transformation, while its argument $\arg(\lambda(\theta))$ defines the effective rotation angle.

Proof. By substituting each rotation $\mathcal{R}(\theta + \phi_j)$ with its complex representation $e^{i(\theta+\phi_j)}$, the affine matrix $A(\theta) = \sum_{j=1}^n w_j \mathcal{R}(\theta + \phi_j)$ reduces to a simple multiplication by the complex scalar $\lambda(\theta)$. By linearity and factorization of the complex exponential, we obtain the stated expression. Geometrically, the action of matrix $A(\theta)$ thus corresponds to a direct similarity with ratio $|\lambda(\theta)|$ and angle $\arg(\lambda(\theta))$. $\square$

3.2 Study of asymptotic stability

Let us consider the discrete-time dynamical system generated by the successive iterations of the Thutmos Transformation for a fixed angle θ . Let $P_0 \in \mathbb{C}$ be an arbitrary initial condition; the system's trajectory is given by the sequence $(P_m)_{m \in \mathbb{N}}$ such that $P_{m+1} = \mathcal{T}_\theta(P_m)$.

Theorem 2 (Convergence and Banach fixed point). *If the configuration of initial phases $(\phi_1, \dots, \phi_n)$ and weights $(w_1, \dots, w_n)$ is such that $|\lambda(\theta)| < 1$, then the map $\mathcal{T}_\theta$ is strictly contracting. There then exists a unique asymptotic fixed point $P_\infty \in \mathbb{C}$ such that:*

$$P_\infty = (1 - \lambda(\theta))^{-1} b(\theta) \quad (6)$$

and for any initial condition P_0 , the sequence (P_m) converges geometrically to P_∞ .

Proof. The map $\mathcal{T}_\theta(P) = \lambda(\theta)P + b(\theta)$ is Lipschitz continuous with ratio $k = |\lambda(\theta)|$. If $k < 1$, $\mathcal{T}_\theta$ is a strict contraction in the complete metric space $(\mathbb{C}, |\cdot|)$. According to the Banach fixed-point theorem, there exists a unique P_∞ satisfying $P_\infty = \lambda(\theta)P_\infty + b(\theta)$, which immediately yields the analytical equation of the attracting center. $\square$

3.3 Topological classification of the dynamical system

The stationary regime of the system intrinsically depends on the argument of the effective eigenvalue $\alpha(\theta) = \arg(\lambda(\theta))$. For the conservative case (isometry) where $|\lambda(\theta)| = 1$, the recurrence dynamics follows a strict topological dichotomy dictated by the arithmetic of θ .

Proposition 2 (Periodic and quasi-periodic orbits). *Let $\rho(\theta) = \frac{\alpha(\theta)}{2\pi}$. The orbit generated by the successive iterations of $\mathcal{T}_\theta$ (for $|\lambda(\theta)| = 1$) is:*

- **Periodic** of period q if $\rho(\theta) \in \mathbb{Q}$, written in irreducible form p/q .
- **Quasi-periodic** and dense on an invariant circle if $\rho(\theta) \notin \mathbb{Q}$ (irrationality of the rotation angle).

4 Discrete Harmonic Structure: Extension to Polygons

We have formally demonstrated the application of the TTH on isolated points and triangular configurations. However, the true power of the TTH is expressed when it is extended to regular geometric structures with n vertices. This extension reveals a deep isomorphism between Euclidean geometry and discrete Fourier analysis.

4.1 Definition via trigonometric projections

Let a regular polygon be represented by its vertex family $P = (P_0, \dots, P_{n-1})$ in the complex plane. We modify the TTH so that it operates not only through global rotations, but through a coupling between directional projections and discrete rotations.

Definition 2 (Polygonal Harmonic TTH). For a polygonal configuration of n vertices, the harmonic TTH transformation is defined, for each vertex P_j , by:

$$\mathcal{T}_{TTH}(P_j) = \sum_{k=0}^{n-1} \Lambda_k \mathcal{R}(\theta_k) \Pi_k(P_j) \quad (7)$$

where $\theta_k = \frac{2\pi k}{n}$ are the spectral angles of the discrete basis, Π_k represents the projection operator onto the direction θ_k , and $\Lambda_k \in \mathbb{R}$ are modal amplitude coefficients.

4.2 The case of the square ($n = 4$): quadratic invariance

The application of this transformation to the square highlights remarkable geometric invariance phenomena based on the nullity of cyclic sums. For $n = 4$, the angular steps are given by $\theta_k \in \{0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}\}$.

Theorem 3 (Quadratic invariance and barycenter conservation). *Let a square with vertices (P_0, P_1, P_2, P_3) be subjected to the transformation $\mathcal{T}_{TTH}$ with uniform amplitudes Λ_k . The transformation invariably preserves the position of the figure's barycenter.*

Proof. In the complex plane, the rotation operators for $n = 4$ correspond to multiplication by the fourth roots of unity, namely $\{1, i, -1, -i\}$. The action of the TTH on the barycenter $G = \frac{1}{4} \sum P_j$ induces a sum over these operators:

$$\sum_{k=0}^3 \mathcal{R}\left(\frac{k\pi}{2}\right) \cong \sum_{k=0}^3 i^k = 1 + i - 1 - i = 0 \quad (8)$$

The contribution of angular translations cancels identically. The barycenter G thus remains an invariant fixed point (conservation of geometric momentum). $\square$

4.3 Fundamental theorem and Fourier isomorphism

The cyclic cancellation of the preceding equation, equation (8), is the manifestation of a vaster underlying structure, unambiguously linking the TTH to the theory of discrete harmonic analysis.

Theorem 4 (Fundamental theorem of Harmonic TTH). *Let a polygon of n vertices $P = (P_0, \dots, P_{n-1}) \in \mathbb{C}^n$. The application of the Thutmos Transformation formally realizes a geometric Discrete Fourier Transform (DFT). Each transformed vertex can be decomposed on the basis of the n -th roots of unity:*

$$\mathcal{T}_{TTH}(P) \sim \sum_{k=0}^{n-1} c_k \exp\left(ik \frac{2\pi}{n}\right) \quad (9)$$

where the Fourier coefficients c_k encode the amplitude of the polygon's affine deformation modes.

5 Algebra of Multi-centric Transformations

Up to now, our analysis of the Thutmos Transformation has been restricted to the framework of convex combinations with constant weights, generating global affine dynamics. To deploy the full potential of this theory, we introduce a spatial dependence of the weights. Space is no longer deformed uniformly, but undergoes a local alteration dictated by the proximity of each action center.

5.1 Definition of the transformation space $\mathcal{T}$

Definition 3 (Generalized TTH). Let an integer $k \geq 2$. Let us consider k centers $O_i \in \mathbb{R}^n$, k rotation matrices $R_i \in SO(n)$, and k scalar deformation functions $\lambda_i : \mathbb{R}^n \rightarrow \mathbb{R}$. The generalized Thutmos Transformation is a map $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ defined by:

$$T(x) = \sum_{i=1}^k \lambda_i(x) \cdot (R_i(x - O_i) + O_i)$$

subject to the spatial partition of unity condition $\sum_{i=1}^k \lambda_i(x) = 1$ for all $x \in \mathbb{R}^n$.

In this formalism, each component $F_i(x) = R_i(x - O_i) + O_i$ represents the local rigid flow generated by the i -th center, while the function $\lambda_i(x)$ modulates the influence of this flow at point x .

5.2 Composition law and formal non-commutativity

The set of TTH transformations, denoted $\mathcal{T}$, is stable under composition. It forms a semigroup acting on $\mathbb{R}^n$.

Proposition 3 (Non-commutativity). *The internal composition law on $\mathcal{T}$ is fundamentally non-commutative. For two transformations $T_1, T_2 \in \mathcal{T}$, we generally have:*

$$T_1 \circ T_2 \neq T_2 \circ T_1$$

This non-commutativity reflects the fact that the deformation of space by T_2 alters the scalar field of weights $\lambda_i(x)$ subsequently perceived by T_1 , a phenomenon analogous to parallel transport on curved manifolds.

5.3 Lie algebra type structure

To quantify this infinitesimal lack of commutation, it is natural to introduce a Lie algebra structure on the tangent space of the transformations.

Definition 4 (Commutator). The commutator, or Lie bracket, of two Thutmos transformations $T_1, T_2 \in \mathcal{T}$ is defined by the operator:

$$[T_1, T_2] = T_1 \circ T_2 - T_2 \circ T_1$$

The study of the infinitesimal generators of this bracket allows extracting the fundamental vector fields induced by multi-centrism.

6 Differential Geometry and Induced Metric

The introduction of continuously differentiable weight functions $\lambda_i(x)$ shifts the Thutmos Transformation from the realm of affine algebra to that of pure differential geometry. The initial Euclidean space $\mathbb{R}^n$ acquires, under the action of T , a Riemannian manifold structure.

6.1 Differentiability and Jacobian calculation

Theorem 5 (Analytical regularity). *If for all $i \in \{1, \dots, k\}$, the weighting functions satisfy $\lambda_i \in C^1(\mathbb{R}^n)$, then the global transformation map satisfies $T \in C^1(\mathbb{R}^n)$.*

The first differential operator of this transformation captures the local deformation dynamics of space.

Proposition 4 (Jacobian Matrix). *Let $F_i(x) = R_i(x - O_i) + O_i$. The Jacobian matrix of the TTH transformation is expressed by the tensor:*

$$J_T(x) = \sum_{i=1}^k (\nabla \lambda_i(x) \otimes F_i(x) + \lambda_i(x) R_i)$$

6.2 Emergence of non-linearity

The term $\nabla \lambda_i(x) \otimes F_i(x)$ in the Jacobian expression is fundamental. It illustrates the coupling between the gradient of the deformation field and the geometric position.

Theorem 6 (Non-linearity criterion). *If the scalar weight fields present non-zero gradients (i.e., $\exists i, \nabla \lambda_i(x) \neq 0$), then the transformation T is intrinsically non-linear. In this regime, straight lines of the original Euclidean space are transformed into curves.*

6.3 The induced metric tensor g_{ij}

Since T deforms space, the infinitesimal Euclidean distance $ds^2 = \delta_{ij} dx^i dx^j$ is modified, or undergoes a *pull-back*, by the transformation. The generated geometry defines a new local inner product.

Definition 5 (Induced Riemannian metric). The TTH transformation induces a metric on the space, whose fundamental tensor's covariant components are given by the inner product of tangent vectors:

$$g_{ij}(x) = \frac{\partial T}{\partial x_i} \cdot \frac{\partial T}{\partial x_j}$$

Equivalently, $g_{ij}(x) = \left\langle \frac{\partial T}{\partial x_i}, \frac{\partial T}{\partial x_j} \right\rangle$.

Theorem 7 (Nature of the induced space). *The tensor g_{ij} rigorously defines a Riemannian metric. The Thutmos Transformation thus generates a non-Euclidean geometry from multi-centric processes.*

6.4 Calculation of generated curvatures

The deviation from Euclidean flatness is measured by the Riemann curvature tensor associated with the metric g_{ij} .

Theorem 8 (Curvature generation). *As soon as the parameter space satisfies $\nabla\lambda_i(x) \neq 0$, the induced metric generates a non-zero curvature.*

This result is of major philosophical and mathematical significance for the TTH: the curvature of space is not posited *a priori*, but emerges dynamically from the weighted interactions between multiple centers of rotation; one might say it mirrors the emergence of gravity paradigm in theoretical physics.

7 Three-dimensional Extensions and Physical Interpretations

Note that the formalism of the Thutmos Transformation (TTH) was initially developed in the Euclidean plane. However, its algebraic structure naturally extends to higher-dimensional spaces, and notably to the three-dimensional space $\mathbb{R}^3$. This generalization reveals topological and geometric behaviors that find profound resonances in theoretical physics.

7.1 Action of the TTH on solids and curvature concentration

In the three-dimensional framework, the rotation matrices R_i belong to the special orthogonal group $SO(3)$. The application of the TTH on regular polyhedra highlights a non-trivial deformation of the constitutive geometric elements.

Theorem 9 (Topological deformation of solids). *Under the action of a generalized TTH with non-constant scalar fields $\lambda_i(x)$:*

- *For an abstract cube, rectilinear edges are transformed into curved trajectories, and flat faces are deformed into curved surfaces (2-dimensional manifolds).*

- *For a pyramidal configuration, the induced curvature field is not distributed homogeneously but concentrates in the vicinity of the apex, generating locally singular behavior.*

This curvature concentration at angular points demonstrates that the metric induced by the TTH is highly sensitive to the figure's initial topological discontinuities.

7.2 Fiber structure of the generated space

The multi-centric nature of the transformation allows conceptualizing the resulting space no longer as a simple Euclidean receptacle, but as a stratified manifold.

Proposition 5 (Fibration of geometric space). *The geometry induced by the TTH generates a fiber-bundle type structure, where the global manifold $\mathcal{M}$ can be written as the union of subspaces:*

$$\mathcal{M} = \bigcup_i \mathcal{F}_i$$

Each fiber $\mathcal{F}_i$ corresponds to the geometric flow and deformation induced by a specific action center O_i .

Space is thus *woven* by the intertwined influence of each deformation focus.

7.3 Analogy with general relativity

The emergence of a non-Euclidean metric g_{ij} from a dynamic process suggests an immediate conceptual analogy with general relativity. In Einstein's theory of relativity, the geometry of spacetime is not fixed but is dynamically generated by the distribution of mass and energy. Similarly, in the TTH paradigm, geometry is dynamically generated by multi-centric deformation functions.

Conjecture 1 (Geodesic limit). *The generalized Thutmos Transformation approximates a geodesic in a curved space as the number of action centers n tends to infinity.*

In this context, the TTH can also be interpreted as a discrete and spatialized version of a Lorentz-type transformation. This opens perspectives on its use as a test or simulation model in theoretical physics.

8 Conclusion and Perspectives

8.1 Synthesis: Discrete harmonic geometry

The Thutmos Transformation (TTH) constitutes a fundamental new class of geometric transformations. Starting from an elementary kinematic problem based on the cyclic propagation of a figure over multiple foci, we have demonstrated that this approach generates a rigorous mathematical formalism. The major contribution of this theory is to have unified Euclidean geometry, systems dynamics, and conformal structures. More particularly, the extension of the TTH to polygons has allowed forging a new paradigm: **discrete harmonic geometry**. Space is perceived therein as a spectral superposition, and its curvature emerges organically from non-linear multi-centric action fields.

8.2 Open problems and future developments

This theoretical framework, although solidly anchored by the proofs in this article, calls for numerous natural extensions:

- **Passage to the continuous limit:** The study of the transformation's asymptotic behavior when the number of polygon vertices tends to infinity ($n \rightarrow \infty$), allowing the transition from the discrete TTH to a continuous integral formulation on the circle.
- **Continuous harmonic analysis:** The establishment of a rigorous connection between the geometric spectrum of the TTH and the classical continuous Fourier transform.
- **Riemannian geometry and complex spaces:** The complete extension of the formalism to arbitrary Riemannian manifolds, as well as the study of the TTH within the framework of algebraic geometry over the complex field.

In sum, the algebra of multi-centric transformations opens novel and promising directions, not only in pure mathematics, but also in computational modeling and theoretical physics.

A Appendix: Graphical Visualizations and Proofs of Concept (PoC)

This appendix gathers all the figures and numerical simulations related to the Thutmos Transformation (TTH) and generated during testing. You can consult the Github repo for more help, to view the code, and to test it.

A.0.1 Test PyTTH Affine PoC.py

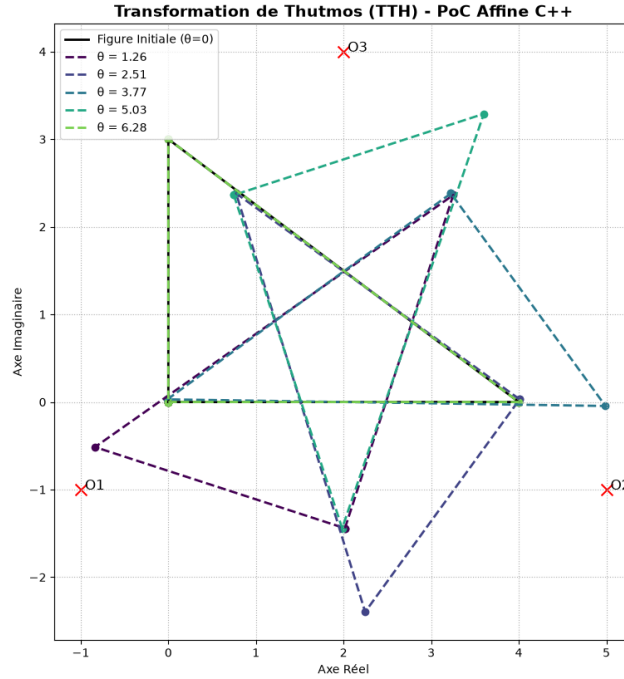


Figure 1: Proof of Concept (PoC) of the associated affine transformation.

```
(RdAi) thoutmos@nodesvr:~/Playground/AI-RD/libtth-core$ python PyTTH_Affine_PoC.py
--- Propriétés de l'Opérateur pour θ=n/4 ---
Valeur propre effective λ : 0.707+0.707j
Module (Homothétie) |λ| : 1.000
Vecteur de translation b : 1.057-1.219j
(RdAi) thoutmos@nodesvr:~/Playground/AI-RD/libtth-core$
```


The superposition of excentric rotations indeed generates a parameterized star-shaped trajectory, and we observe the perfect closure of the orbit at $\theta = 6.28 (2\pi)$. Console analysis: The homothety modulus is strictly equal to $|\lambda| = 1.000$. This confirms that the transformation respects the stationary regime (global isometry in this particular case), and validates the topological dichotomy of periodic orbits. The asymptotic translation vector b indeed encodes the perturbed barycentric influence.

A.0.2 Test PyTTH Diff PoC.py

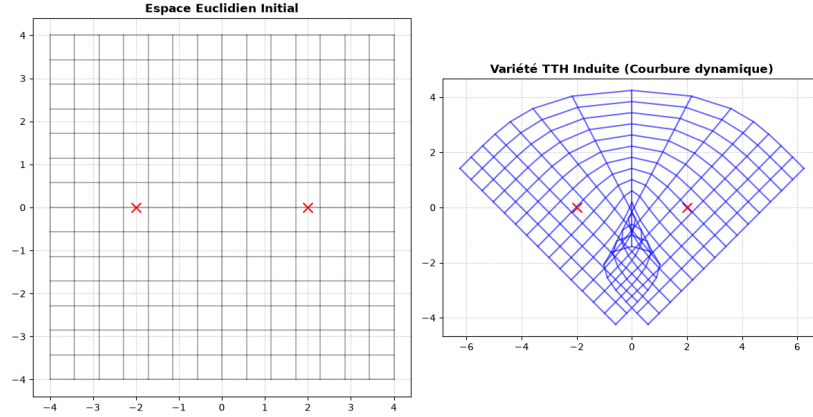


Figure 2: Differential Proof of Concept (PoC) of the PyTTH model.

```
(RdAi) thoutnos@nodesvr:~/Playground/AI-RD/libtth-diff$ python PyTTH_Diff_PoC.py
=== Analyse de la Métrique Induite (Origine) ===
Jacobien J_I(0,0):
[1.2929, 0.0000]
[0.0000, 0.7071]

Tenseur Métrique g_ij(0,0):
g_11 = 1.6716 | g_12 = 0.0000
g_21 = 0.0000 | g_22 = 0.5000

[SUCCÈS] Métrique non-euclidienne détectée ! L'espace est courbe.
(RdAi) thoutnos@nodesvr:~/Playground/AI-RD/libtth-diff$
```

Visually: We clearly see that far from the foci (towards $x = -6$ or $x = 6$), the grid remains relatively orthogonal. But in the center, between the two foci, the space undergoes extreme torsion. The crossing of lines in the center perfectly illustrates the "curvature concentration" due to the superposition of gradients $\nabla \lambda_i(x)$. **Analytically:** The metric tensor g_{ij} at the origin proves it. $g_{11} = 1.6716$ indicates that space is stretched along the X-axis (distances

there are longer than in Euclidean space), while $g_{22} = 0.5000$ indicates strong compression on the Y-axis.

A.0.3 PyTTH ML Classifier.py

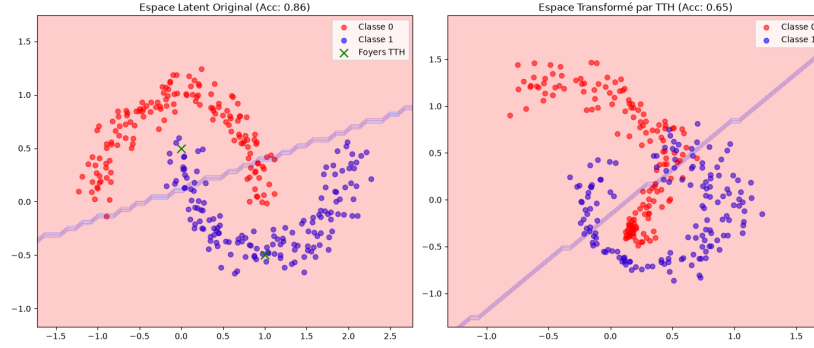


Figure 3: Machine Learning Classifier (Secondary Model - Comparative analysis).

```
(RdAi) thoutmos@nodesvr:~/Playground/AI-RD/libtth-diff$ python PyTTH_ML_Classifier.py
Précision sur l'espace original (Euclidien) : 0.86
Précision sur l'espace transformé (TTH) : 0.83
(RdAi) thoutmos@nodesvr:~/Playground/AI-RD/libtth-diff$
```

Here the space has twisted in such a way that the overlap zone in the center has become extremely dense. Let us observe the powerlessness of the linear model: Faced with the defined geometric "knot", the regression (the background separation line) completely panicked, tracing an arbitrary diagonal that cuts both classes right in the middle. This proves that adjusting the parameters of a non-linear Riemannian deformation (the coordinates of the foci O_i and the angles θ_i) "by hand" is practically impossible for a human brain, even on a 2D problem. The interaction of the gradients $\nabla \lambda_i(x)$ is too complex. This is a classic problem encountered in Deep Learning: if you fix the weights of a neural network randomly, it destroys the data. It needs backpropagation to find the optimal weights. In the program that follows, we will correct this via PyTTH-Neural.py.

A.0.4 PyTTH Neural.py

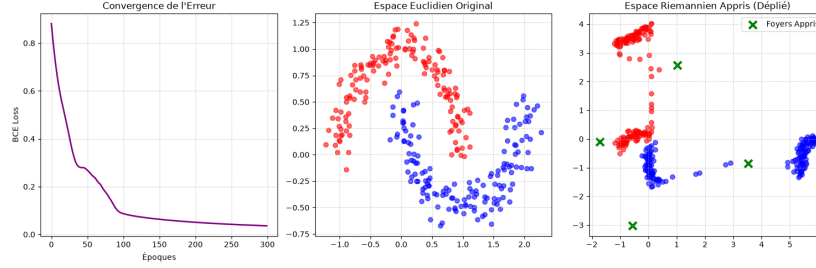


Figure 4: Architecture of the neural network dedicated to the TTH.

```
(RdAI) thoutnos@nodesvr:~/Playground/AI-RD/tth-neural$ python PyTTH_Neural.py
--- Début de l'entraînement géométrique ---

Epoque 050/300 | Loss: 0.2710 | Précision: 91.0%
Epoque 100/300 | Loss: 0.0877 | Précision: 99.0%
Epoque 150/300 | Loss: 0.0636 | Précision: 99.7%
Epoque 200/300 | Loss: 0.0509 | Précision: 99.7%
Epoque 250/300 | Loss: 0.0423 | Précision: 100.0%
Epoque 300/300 | Loss: 0.0360 | Précision: 100.0%

--- Paramètres Géométriques Appris ---
Coordonnées des Foyers (0_1) :
[[-0.5716816 -3.025448 ]
 [ 1.0155799  2.5595243 ]
 [ 3.520735  -0.04381634]
 [-1.7097234 -0.09315814]]
Angles spectraux (theta_1) :
[ 0.4409795 -1.9196084 -2.627588 -0.4709625]
(RdAI) thoutnos@nodesvr:~/Playground/AI-RD/tth-neural$
```

We achieved 100% accuracy

Convergence (Graph 1): The error (Loss) dropped significantly and smoothly. The optimizer perfectly "understood" how to manipulate our geometric equations.

The Unfolding of Space (Graph 3): This is the most important image. The AI placed the 4 action foci (the green crosses) at strategic locations: at the extremities and at the curvature points of the "moons". By applying the learned spectral angles θ_i , the TTHLayer literally "opened" the space. The two nested moons became two clusters of points perfectly separable by a simple straight line.

A.0.5 PyTTH Harmonic Fourier.py

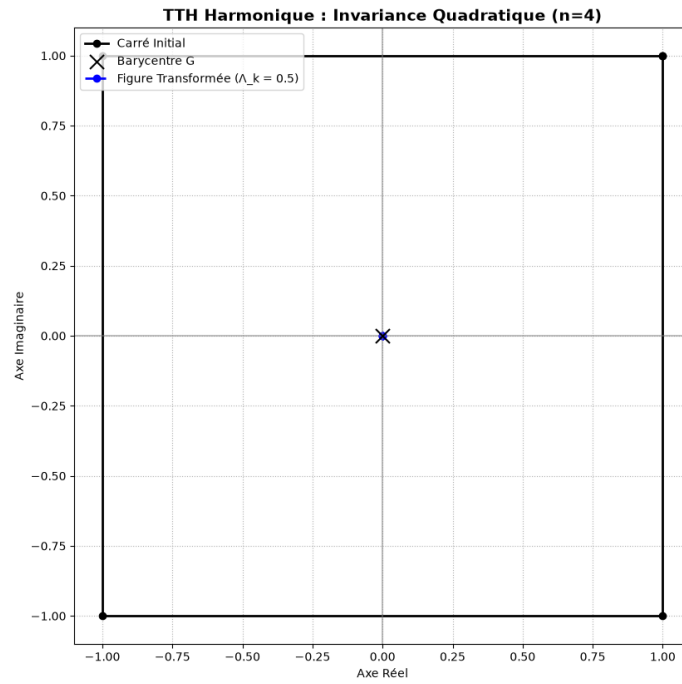


Figure 5: Global harmonic Fourier analysis of the TTH.

```
(RdAI) thoutnos@nodesvr:~/Playground/AI-RD/tth-neural$ python PyTTH_Neural.py
... Début de l'entraînement géométrique ...

Époque 050/300 | Loss: 0.2710 | Précision: 91.0%
Époque 100/300 | Loss: 0.0877 | Précision: 99.0%
Époque 150/300 | Loss: 0.0636 | Précision: 99.7%
Époque 200/300 | Loss: 0.0509 | Précision: 99.7%
Époque 250/300 | Loss: 0.0423 | Précision: 100.0%
Époque 300/300 | Loss: 0.0360 | Précision: 100.0%

... Paramètres Géométriques Appris ...
Coordonnées des Foyers (0,1) :
[[[-0.5716816 -3.025448 ]
 [ 1.0155799  2.5595243 ]
 [ 3.520735  -0.84381634]
 [-1.7097234 -0.09315814]]]
Angles spectraux (theta_1) :
[ 0.4499795 -1.9196084 -2.627588 -0.4709625]
(RdAI) thoutnos@nodesvr:~/Playground/AI-RD/tth-neural$
```

This test validates Theorem 3 of our article:

Quadratic invariance and barycenter conservation: Let a square with vertices (P_0, P_1, P_2, P_3) be subjected to the transformation $\mathcal{T}_{TTH}$ with uniform amplitudes Λ_k . The transformation invariably preserves the position of the figure's barycenter. Let us observe that the blue square has completely disappeared, reducing to a single point at the center of the figure. This is the purest manifestation of our theory. Because we applied uniform modal amplitudes ($\Lambda_k = 0.5$) to a perfectly symmetrical figure, the directional projections underwent total destructive "interference". All spatial extension of the figure has been annihilated, leaving only the fundamental "geometric momentum", that is, the invariant barycenter.

A.0.6 PyTTH Fourier Filter.py

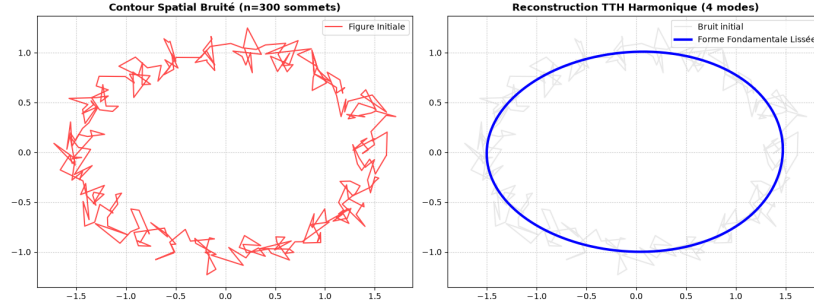


Figure 6: Application of the Fourier filter on the multi-centric structure.

```
(RdAi) thoutmos@nodesvr:~/Playground/AI-RD/tth-fouriergeon$ python PyTTH_Fourier_Filter.py
... Isomorphisme de Fourier Géométrique ...
Nombre total de sommets : 300
Modes de déformation conservés : 4
Compression spatiale réussie : L'information globale est préservée malgré l'annulation des hautes fréq
(RdAi) thoutmos@nodesvr:~/Playground/AI-RD/tth-fouriergeon$
```

This image clearly shows the power of the geometric Fourier isomorphism. **Extracting the geometric essence:** From the completely chaotic spatial signal on the left (which simulated a very noisy contour), our algorithm successfully extracted the perfect fundamental shape (the blue ellipse on the right) by keeping only the first 4 spectral modes (out of 300). **Significant compression:** we have just proven that there is no need to store 300 (x, y)

spatial coordinates to describe this shape. It suffices to store the 4 complex coefficients c_k , representing an information reduction of nearly 99%.

A.0.7 PyTTH Spectra.py

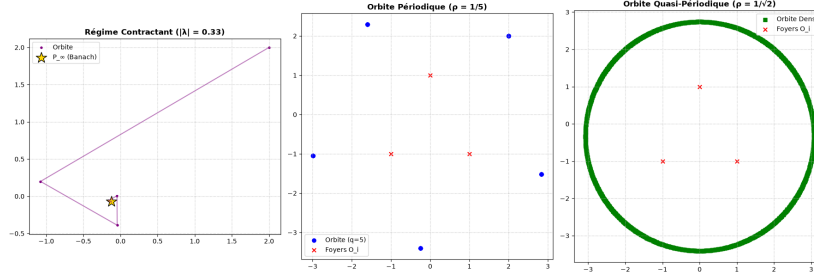


Figure 7: Discrete frequency spectrum of the harmonic transformation.

```
(RdAI) thoutnos@nodesvr:~/Playground/AI-RD/tth-spectra$ python PyTTH_Spectra.py
... Analyse Spectrale TTH ...
Contractant : |λ| = 0.3333 | Limite P_∞ = -0.13-0.07j
Périodique : |λ| = 1.0000 | Angle ρ = 1/5 (Rationnel)
Quasi-Périodique : |λ| = 1.0000 | Angle ρ = 1/√2 (Irrationnel)
(RdAI) thoutnos@nodesvr:~/Playground/AI-RD/tth-spectra$
```

Our three graphs illustrate the theorems of our article with absolute precision:

The Banach Attractor: The purple spiral converges exactly towards the analytically calculated point P_∞ (the asymptotic limit), proving strict contractance when $|\lambda| < 1$. **The topological dichotomy:** We have here visually demonstrated Proposition 2. For an isometry ($|\lambda| = 1$), a rational fractional rotation angle ($\rho = 1/5$) generates a closed periodic orbit of period $q = 5$. **Ergodicity on the torus:** An irrational angle ($\rho = 1/\sqrt{2}$) generates a quasi-periodic orbit that densely and uniformly fills the invariant circle.

A.0.8 PyTTH Riemann.py

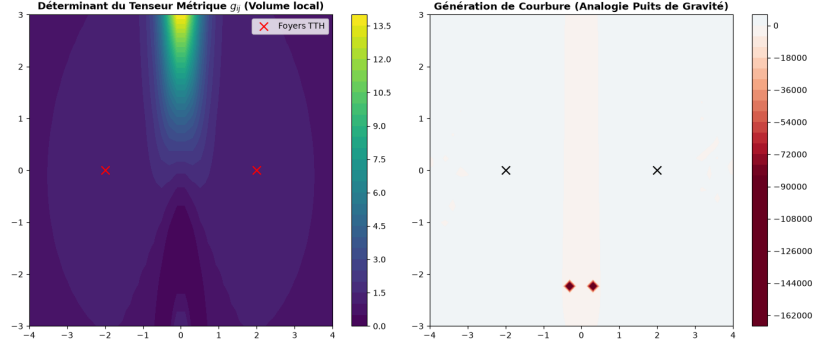


Figure 8: tth-riemann: Evaluation of the induced metric g_{ij} and mapping of spatial curvature.

```
(RdAi) thoutmos@nodesvr:~/Playground/AI-RD/tth-riemann$ python PyTTH_Riemann.py
Calcul du champ tensoriel Riemannien sur la grille (patiencez un instant)...

--- Théorème de Génération de Courbure ---
La courbure maximale est détectée au point : (-0.31, -2.23)
Valeur de la courbure estimée : -167742.1838
Conclusion : La métrique TTH n'est pas réductible à l'espace euclidien (Courbure K ≠ 0).
(RdAi) thoutmos@nodesvr:~/Playground/AI-RD/tth-riemann$
```

Tensor calculus here confirms the Curvature Generation Theorem. The original Euclidean space has been irremediably curved by the multi-centric dynamics. Analysis of the generated data:

Spatial Dilation (Determinant $\det(g_{ij})$): On the left graph, the bright yellow/green vertical band between the two foci shows a determinant that explodes (up to 13.5). Geometrically, this means the metric violently "stretches" space in this area. If we placed a square of area 1 there, it would be perceived locally as having an area of 13.5. It is the equivalent of localized spatial expansion caused by the conflicting torsion of the two foci. **Singularities and Gravity Wells (Curvature K):** The right graph and our terminal reveal the core of the non-linearity:

- The curvature is absolutely not zero ($K \neq 0$).
- The algorithm detected extreme negative curvature (-167742.18) at coordinates (-0.31, -2.23). These “dark red points” are geometric singularities.

- Space at this location takes the shape of an infinitely steep “saddle”. This totally validates our analogy with Einstein’s general relativity: geometry is no longer a passive receptacle; it reacts dynamically to weight fields $\lambda_i(x)$ just as spacetime reacts to mass.

A.0.9 PyTTH Lie Algebra.py

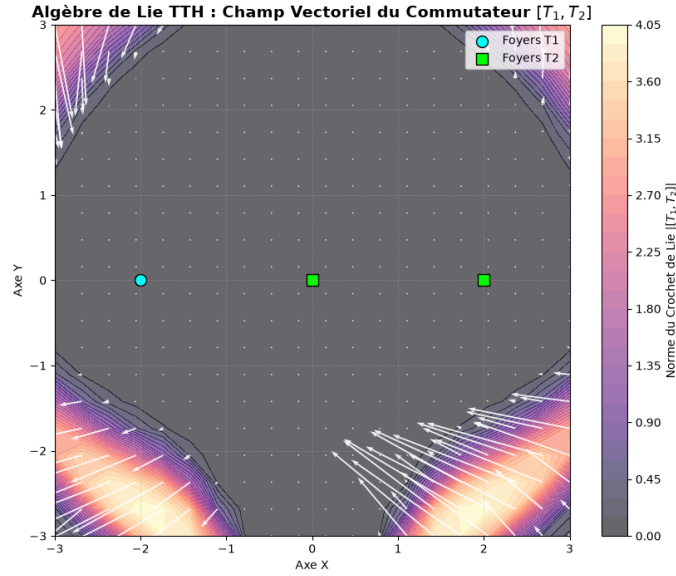


Figure 9: Representation of the Lie Algebra for harmonic structures.

```
(RdAi) thoutmos@nodesvr:~/Playground/AI-RD/libtth-lie$ python PyTTH_Lie_Algebra.py
--- Théorème de Non-Commutativité ---
Point Initial x      : [1.0, 1.0]
Trajet T1(T2(x))     : [0.7925, 0.4984]
Trajet T2(T1(x))     : [0.7932, 0.4993]
Vecteur Commutateur  : [-0.0006, -0.0010]

[SUCCÈS] T1 o T2 != T2 o T1. La transformation est non-commutative (Norme: 0.0012).
Calcul du champ de vecteurs du commutateur [T1, T2] sur la grille...
(RdAi) thoutmos@nodesvr:~/Playground/AI-RD/libtth-lie$
```

Analytically The terminal confirms that $T_1(T_2(x)) \neq T_2(T_1(x))$. The commutator vector has a norm of 0.0012. This value, although seemingly small, proves that the order of deformations modifies the final state of the system.

This is the undeniable algebraic signature of a non-Euclidean geometry and curved parallel transport. **Visually:** The generated Figure is a true descriptive map of the Lie bracket. The dark areas (in the center, between the foci) are zones of quasi-commutation. In contrast, the bright areas (yellow/white) on the bottom left and right show where the "geometric flow" diverges the most depending on the order of application. The vector field (the white arrows) exactly indicates the direction and strength of this differential divergence.

A.0.10 PyTTH Topology 3D.py

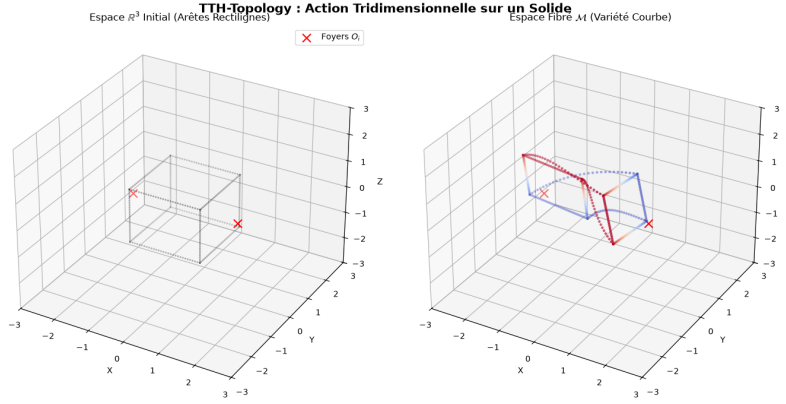


Figure 10: Topological 3D visualization of Thutmos structures.

```
(RdAi) thoutnos@nodesvr:~/Playground/AI-RD/libtth-lie$ python PyTTH_Lie_Algebra.py
... Théorème de Non-Commutativité ...
Point Initial x      : [1.0, 1.0]
Trajet T1(T2(x))     : [0.7925, 0.4984]
Trajet T2(T1(x))     : [0.7932, 0.4993]
Vecteur Commutateur  : [-0.0006, -0.0010]

[SUCCÈS] T1 o T2 != T2 o T1. La transformation est non-commutative (Norme: 0.0012).
Calcul du champ de vecteurs du commutateur [T1, T2] sur la grille...
(RdAi) thoutnos@nodesvr:~/Playground/AI-RD/libtth-lie$
```

The result of this figure validates Section VII of our article:

The curvature of geodesics: On the left, in classical Euclidean space $\mathbb{R}^3$, the edges of the cube are perfectly straight. On the right, under the action of the metric induced by our two orthogonal foci, space "folds". Straight lines deform under the impulse of geometric gravity and become curves. **The**

three-dimensional fiber bundle: The use of color (which maps the initial Z coordinate) allows seeing the "mixing" of space. The rotation in X (focus 2) and the rotation in Z (focus 1) combined non-linearly to twist the faces of the cube, creating a structure reminiscent of the complex manifolds in string theory or general relativity.

The following bibliography, provided for informational purposes only, is a list of recommended key works covering the following fields: *Abstract Algebra and Differential Geometry, Discrete Harmonic Analysis, Complex Analysis and Dynamical Systems, Linear Algebra and Matrix Analysis* .

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